

# Phase I/II Operating Characteristics

New non-TITE alpha comparison with cycle-1 oracle (alpha = 0)

|                                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No OBD % |
|------------------------------------------------------|--------|--------|--------|--------|--------|----------|
| <b>Scenario 1</b>                                    |        |        |        |        |        |          |
| DLT rate                                             | 0.3    | 0.35   | 0.4    | 0.45   | 0.5    |          |
| Efficacy rate                                        | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |          |
| <b>OBD selection (%)</b>                             |        |        |        |        |        |          |
| Oracle   one-stage   cycle 1, alpha=0                | 16.9   | 5.4    | 1.8    | 0.4    | 0.1    | 75.4     |
| New non-TITE   one-stage   alpha=0                   | 17     | 7.2    | 1.7    | 0.1    | 0.1    | 73.9     |
| New non-TITE   one-stage   alpha=0.3                 | 15.7   | 3.6    | 1.2    | 0.1    | 0.1    | 79.3     |
| New non-TITE   one-stage   alpha=0.6                 | 12.5   | 2.5    | 0.5    | 0.1    | 0      | 84.4     |
| New non-TITE   one-stage   alpha=0.9                 | 11     | 1.1    | 0.3    | 0      | 0      | 87.6     |
| Oracle   2-stage highest utility   cycle 1, alpha=0  | 17.6   | 5.7    | 0.9    | 0.4    | 0      | 75.4     |
| New non-TITE   2-stage highest utility   alpha=0     | 18     | 7.2    | 1.2    | 0      | 0.2    | 73.4     |
| New non-TITE   2-stage highest utility   alpha=0.3   | 15.7   | 4.2    | 0.7    | 0      | 0.1    | 79.3     |
| New non-TITE   2-stage highest utility   alpha=0.6   | 13     | 2.2    | 0.3    | 0      | 0      | 84.5     |
| New non-TITE   2-stage highest utility   alpha=0.9   | 10.9   | 1.4    | 0.3    | 0      | 0      | 87.4     |
| Oracle   2-stage top-2 randomized   cycle 1, alpha=0 | 16.6   | 5.7    | 1.5    | 0.5    | 0.1    | 75.6     |
| New non-TITE   2-stage top-2 randomized   alpha=0    | 17.4   | 7.2    | 2      | 0.8    | 0.1    | 72.5     |
| New non-TITE   2-stage top-2 randomized   alpha=0.3  | 14.9   | 5.2    | 0.9    | 0.2    | 0      | 78.8     |
| New non-TITE   2-stage top-2 randomized   alpha=0.6  | 12.1   | 2.6    | 0.3    | 0      | 0      | 85       |
| New non-TITE   2-stage top-2 randomized   alpha=0.9  | 10.6   | 2.1    | 0      | 0      | 0      | 87.3     |
| <b>Mean administrations by dose</b>                  |        |        |        |        |        |          |
| Oracle   one-stage   cycle 1, alpha=0                | 11.1   | 4.3    | 2.2    | 0.8    | 0.2    |          |
| New non-TITE   one-stage   alpha=0                   | 11.1   | 4.4    | 2.3    | 0.9    | 0.3    |          |
| New non-TITE   one-stage   alpha=0.3                 | 11.5   | 3.9    | 1.7    | 0.6    | 0.2    |          |
| New non-TITE   one-stage   alpha=0.6                 | 11.7   | 3.4    | 1.2    | 0.3    | 0.1    |          |
| New non-TITE   one-stage   alpha=0.9                 | 12     | 2.6    | 0.8    | 0.2    | 0      |          |
| Oracle   2-stage highest utility   cycle 1, alpha=0  | 11.2   | 4.2    | 2.3    | 0.8    | 0.2    |          |
| New non-TITE   2-stage highest utility   alpha=0     | 11.1   | 4.4    | 2.3    | 1      | 0.3    |          |
| New non-TITE   2-stage highest utility   alpha=0.3   | 11.4   | 3.9    | 1.8    | 0.6    | 0.1    |          |
| New non-TITE   2-stage highest utility   alpha=0.6   | 11.7   | 3.3    | 1.3    | 0.3    | 0.1    |          |
| New non-TITE   2-stage highest utility   alpha=0.9   | 11.9   | 2.6    | 0.9    | 0.2    | 0      |          |
| Oracle   2-stage top-2 randomized   cycle 1, alpha=0 | 11.1   | 4.2    | 2.2    | 0.9    | 0.2    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0    | 11     | 4.4    | 2.3    | 1      | 0.3    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.3  | 11.3   | 3.9    | 1.7    | 0.6    | 0.1    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.6  | 11.7   | 3.3    | 1.2    | 0.3    | 0.1    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.9  | 11.9   | 2.7    | 0.9    | 0.2    | 0      |          |

# Phase I/II Operating Characteristics

New non-TITE alpha comparison with cycle-1 oracle (alpha = 0)

|                                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No OBD % |
|------------------------------------------------------|--------|--------|--------|--------|--------|----------|
| <b>Scenario 16</b>                                   |        |        |        |        |        |          |
| DLT rate                                             | 0.02   | 0.06   | 0.09   | 0.14   | 0.26   |          |
| Efficacy rate                                        | 0.2    | 0.2    | 0.2    | 0.2    | 0.2    |          |
| <b>OBD selection (%)</b>                             |        |        |        |        |        |          |
| Oracle   one-stage   cycle 1, alpha=0                | 29.9   | 20.9   | 20.4   | 18.5   | 10.3   | 0        |
| New non-TITE   one-stage   alpha=0                   | 29.3   | 21.3   | 20.2   | 18.8   | 10.4   | 0        |
| New non-TITE   one-stage   alpha=0.3                 | 29.6   | 24.3   | 18.5   | 17.3   | 10.3   | 0        |
| New non-TITE   one-stage   alpha=0.6                 | 26.7   | 27.2   | 19.1   | 16.9   | 10.1   | 0        |
| New non-TITE   one-stage   alpha=0.9                 | 24.9   | 29.8   | 19.2   | 16.9   | 9.2    | 0        |
| Oracle   2-stage highest utility   cycle 1, alpha=0  | 32.5   | 22.7   | 16.3   | 17.8   | 10.6   | 0.1      |
| New non-TITE   2-stage highest utility   alpha=0     | 34.1   | 21.3   | 16.6   | 16.4   | 11.5   | 0.1      |
| New non-TITE   2-stage highest utility   alpha=0.3   | 30.9   | 26.8   | 18.6   | 16.3   | 7.4    | 0        |
| New non-TITE   2-stage highest utility   alpha=0.6   | 29.4   | 28.3   | 18     | 16.1   | 8.2    | 0        |
| New non-TITE   2-stage highest utility   alpha=0.9   | 27     | 33.9   | 16.9   | 15.6   | 6.6    | 0        |
| Oracle   2-stage top-2 randomized   cycle 1, alpha=0 | 31.5   | 22.3   | 17.1   | 16.1   | 13     | 0        |
| New non-TITE   2-stage top-2 randomized   alpha=0    | 33.9   | 22.2   | 16.5   | 15.4   | 12     | 0        |
| New non-TITE   2-stage top-2 randomized   alpha=0.3  | 32.1   | 23.8   | 17.6   | 15.8   | 10.7   | 0        |
| New non-TITE   2-stage top-2 randomized   alpha=0.6  | 29.5   | 26.5   | 18.9   | 14.8   | 10.3   | 0        |
| New non-TITE   2-stage top-2 randomized   alpha=0.9  | 26.4   | 29.8   | 17.6   | 18     | 8.2    | 0        |
| <b>Mean administrations by dose</b>                  |        |        |        |        |        |          |
| Oracle   one-stage   cycle 1, alpha=0                | 7.3    | 6.7    | 5.8    | 5.7    | 4.5    |          |
| New non-TITE   one-stage   alpha=0                   | 7.3    | 6.7    | 5.7    | 5.7    | 4.6    |          |
| New non-TITE   one-stage   alpha=0.3                 | 7.3    | 7      | 5.7    | 5.7    | 4.3    |          |
| New non-TITE   one-stage   alpha=0.6                 | 7.2    | 7.5    | 5.8    | 5.5    | 4.1    |          |
| New non-TITE   one-stage   alpha=0.9                 | 7.1    | 8.1    | 5.9    | 5.3    | 3.6    |          |
| Oracle   2-stage highest utility   cycle 1, alpha=0  | 7.5    | 6.9    | 5.8    | 5.6    | 4.2    |          |
| New non-TITE   2-stage highest utility   alpha=0     | 7.6    | 6.8    | 5.7    | 5.5    | 4.4    |          |
| New non-TITE   2-stage highest utility   alpha=0.3   | 7.4    | 7.3    | 5.7    | 5.5    | 4.1    |          |
| New non-TITE   2-stage highest utility   alpha=0.6   | 7.2    | 7.8    | 5.7    | 5.4    | 3.9    |          |
| New non-TITE   2-stage highest utility   alpha=0.9   | 7.2    | 8.2    | 5.5    | 5.3    | 3.7    |          |
| Oracle   2-stage top-2 randomized   cycle 1, alpha=0 | 6.6    | 6.3    | 5.7    | 6.1    | 5.3    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0    | 6.6    | 6.1    | 5.6    | 6.1    | 5.6    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.3  | 6.6    | 6.5    | 5.7    | 6      | 5.2    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.6  | 6.5    | 6.7    | 5.8    | 6      | 4.9    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.9  | 6.5    | 7.1    | 5.8    | 6.1    | 4.6    |          |

# Phase I/II Operating Characteristics

New non-TITE alpha comparison with cycle-1 oracle (alpha = 0)

|                                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No OBD % |
|------------------------------------------------------|--------|--------|--------|--------|--------|----------|
| <b>Scenario 20</b>                                   |        |        |        |        |        |          |
| DLT rate                                             | 0.07   | 0.14   | 0.31   | 0.41   | 0.52   |          |
| Efficacy rate                                        | 0.05   | 0.25   | 0.25   | 0.25   | 0.25   |          |
| <b>OBD selection (%)</b>                             |        |        |        |        |        |          |
| Oracle   one-stage   cycle 1, alpha=0                | 13.5   | 46.2   | 24.6   | 11     | 1.5    | 3.2      |
| New non-TITE   one-stage   alpha=0                   | 14.4   | 43.9   | 25.7   | 10.8   | 2.4    | 2.8      |
| New non-TITE   one-stage   alpha=0.3                 | 12.9   | 48.4   | 25.1   | 8.5    | 1.4    | 3.7      |
| New non-TITE   one-stage   alpha=0.6                 | 10.9   | 50.9   | 26.2   | 6.9    | 0.8    | 4.3      |
| New non-TITE   one-stage   alpha=0.9                 | 9.4    | 54.8   | 24.6   | 6      | 0.4    | 4.8      |
| Oracle   2-stage highest utility   cycle 1, alpha=0  | 13.5   | 51.2   | 24.4   | 6.6    | 0.7    | 3.6      |
| New non-TITE   2-stage highest utility   alpha=0     | 14.2   | 47.4   | 25.8   | 7.7    | 1.7    | 3.2      |
| New non-TITE   2-stage highest utility   alpha=0.3   | 11.8   | 53.6   | 23.6   | 6.1    | 0.8    | 4.1      |
| New non-TITE   2-stage highest utility   alpha=0.6   | 10.2   | 55.4   | 24.2   | 5.4    | 0.4    | 4.4      |
| New non-TITE   2-stage highest utility   alpha=0.9   | 10     | 57     | 23.5   | 4.1    | 0.2    | 5.2      |
| Oracle   2-stage top-2 randomized   cycle 1, alpha=0 | 9.2    | 50     | 25.3   | 9.8    | 2      | 3.7      |
| New non-TITE   2-stage top-2 randomized   alpha=0    | 10.4   | 47.7   | 25.6   | 10.6   | 2.7    | 3        |
| New non-TITE   2-stage top-2 randomized   alpha=0.3  | 8.2    | 52.8   | 24.4   | 9.2    | 1.7    | 3.7      |
| New non-TITE   2-stage top-2 randomized   alpha=0.6  | 9.7    | 55.5   | 21.4   | 7.6    | 1.1    | 4.7      |
| New non-TITE   2-stage top-2 randomized   alpha=0.9  | 8.8    | 56.7   | 24.1   | 4.7    | 0.6    | 5.1      |
| <b>Mean administrations by dose</b>                  |        |        |        |        |        |          |
| Oracle   one-stage   cycle 1, alpha=0                | 5.2    | 11.3   | 7.6    | 4      | 1.2    |          |
| New non-TITE   one-stage   alpha=0                   | 5.2    | 11     | 7.6    | 4.2    | 1.4    |          |
| New non-TITE   one-stage   alpha=0.3                 | 5.5    | 11.6   | 7.5    | 3.8    | 1      |          |
| New non-TITE   one-stage   alpha=0.6                 | 5.7    | 12.1   | 7.4    | 3.4    | 0.7    |          |
| New non-TITE   one-stage   alpha=0.9                 | 5.8    | 12.7   | 7.3    | 2.9    | 0.5    |          |
| Oracle   2-stage highest utility   cycle 1, alpha=0  | 5.4    | 11.6   | 8      | 3.5    | 1      |          |
| New non-TITE   2-stage highest utility   alpha=0     | 5.3    | 11.2   | 7.9    | 3.8    | 1.3    |          |
| New non-TITE   2-stage highest utility   alpha=0.3   | 5.6    | 11.6   | 7.8    | 3.3    | 1      |          |
| New non-TITE   2-stage highest utility   alpha=0.6   | 5.7    | 12.1   | 7.7    | 3      | 0.8    |          |
| New non-TITE   2-stage highest utility   alpha=0.9   | 6      | 12.5   | 7.4    | 2.7    | 0.7    |          |
| Oracle   2-stage top-2 randomized   cycle 1, alpha=0 | 5.7    | 9.8    | 8      | 4.4    | 1.5    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0    | 5.7    | 9.5    | 8      | 4.6    | 1.8    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.3  | 6      | 9.9    | 8      | 4.2    | 1.3    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.6  | 6.3    | 10.3   | 7.9    | 3.8    | 1.1    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.9  | 6.7    | 10.6   | 7.9    | 3.3    | 0.8    |          |

# Phase I/II Operating Characteristics

New non-TITE alpha comparison with cycle-1 oracle (alpha = 0)

|                                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No OBD % |
|------------------------------------------------------|--------|--------|--------|--------|--------|----------|
| <b>Scenario 24</b>                                   |        |        |        |        |        |          |
| DLT rate                                             | 0.01   | 0.02   | 0.05   | 0.1    | 0.25   |          |
| Efficacy rate                                        | 0.05   | 0.08   | 0.2    | 0.5    | 0.5    |          |
| <b>OBD selection (%)</b>                             |        |        |        |        |        |          |
| Oracle   one-stage   cycle 1, alpha=0                | 1.5    | 1.3    | 5.5    | 59.6   | 32.1   | 0        |
| New non-TITE   one-stage   alpha=0                   | 1.4    | 0.9    | 5      | 58.6   | 34.1   | 0        |
| New non-TITE   one-stage   alpha=0.3                 | 0.8    | 0.9    | 3.4    | 61     | 33.9   | 0        |
| New non-TITE   one-stage   alpha=0.6                 | 0.5    | 1.1    | 4.2    | 64.4   | 29.8   | 0        |
| New non-TITE   one-stage   alpha=0.9                 | 0.5    | 0.7    | 3.2    | 66.6   | 29     | 0        |
| Oracle   2-stage highest utility   cycle 1, alpha=0  | 2      | 1.1    | 6.8    | 56.7   | 33.4   | 0        |
| New non-TITE   2-stage highest utility   alpha=0     | 2      | 1.5    | 6.2    | 55.5   | 34.8   | 0        |
| New non-TITE   2-stage highest utility   alpha=0.3   | 1.1    | 1.4    | 5.5    | 55.9   | 36.1   | 0        |
| New non-TITE   2-stage highest utility   alpha=0.6   | 1.2    | 0.9    | 5.6    | 56.3   | 36     | 0        |
| New non-TITE   2-stage highest utility   alpha=0.9   | 0.8    | 0.8    | 5.3    | 57.3   | 35.8   | 0        |
| Oracle   2-stage top-2 randomized   cycle 1, alpha=0 | 0.3    | 0.3    | 2.9    | 56.3   | 40.2   | 0        |
| New non-TITE   2-stage top-2 randomized   alpha=0    | 0.4    | 0.8    | 3.6    | 56.6   | 38.6   | 0        |
| New non-TITE   2-stage top-2 randomized   alpha=0.3  | 0.8    | 0.5    | 4      | 56.1   | 38.6   | 0        |
| New non-TITE   2-stage top-2 randomized   alpha=0.6  | 0.6    | 0.6    | 4.2    | 54.6   | 40     | 0        |
| New non-TITE   2-stage top-2 randomized   alpha=0.9  | 0.9    | 0.8    | 2.9    | 55.9   | 39.5   | 0        |
| <b>Mean administrations by dose</b>                  |        |        |        |        |        |          |
| Oracle   one-stage   cycle 1, alpha=0                | 3.4    | 3.5    | 4.5    | 12.5   | 6.1    |          |
| New non-TITE   one-stage   alpha=0                   | 3.4    | 3.5    | 4.4    | 12.2   | 6.5    |          |
| New non-TITE   one-stage   alpha=0.3                 | 3.4    | 3.4    | 4.3    | 12.9   | 6      |          |
| New non-TITE   one-stage   alpha=0.6                 | 3.4    | 3.3    | 4.3    | 13.6   | 5.4    |          |
| New non-TITE   one-stage   alpha=0.9                 | 3.3    | 3.4    | 4.2    | 14     | 5.1    |          |
| Oracle   2-stage highest utility   cycle 1, alpha=0  | 3.4    | 3.5    | 4.4    | 10.9   | 7.7    |          |
| New non-TITE   2-stage highest utility   alpha=0     | 3.4    | 3.5    | 4.4    | 10.6   | 8.1    |          |
| New non-TITE   2-stage highest utility   alpha=0.3   | 3.4    | 3.4    | 4.4    | 10.8   | 8      |          |
| New non-TITE   2-stage highest utility   alpha=0.6   | 3.4    | 3.4    | 4.2    | 11     | 8      |          |
| New non-TITE   2-stage highest utility   alpha=0.9   | 3.4    | 3.4    | 4.1    | 11.1   | 8      |          |
| Oracle   2-stage top-2 randomized   cycle 1, alpha=0 | 3.6    | 3.8    | 4.9    | 9.5    | 8.4    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0    | 3.6    | 3.7    | 4.9    | 9.4    | 8.4    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.3  | 3.5    | 3.7    | 4.8    | 9.4    | 8.5    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.6  | 3.5    | 3.8    | 4.8    | 9.5    | 8.5    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.9  | 3.5    | 3.8    | 4.8    | 9.5    | 8.4    |          |

Sources: cycle-2 new-design alpha grid; cycle-1 continuous one-stage oracle; existing two-stage oracle rows (1,000 simulations per populated row).

Oracle rows use Cycle\_Max = 1 and alpha = 0; No OBD % is the source's no-OBD selection percentage.

# Phase I/II Operating Characteristics

New non-TITE alpha comparison with cycle-1 oracle (alpha = 0)

|                                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No OBD % |
|------------------------------------------------------|--------|--------|--------|--------|--------|----------|
| <b>Scenario 27</b>                                   |        |        |        |        |        |          |
| DLT rate                                             | 0.11   | 0.17   | 0.3    | 0.5    | 0.62   |          |
| Efficacy rate                                        | 0.1    | 0.2    | 0.4    | 0.42   | 0.45   |          |
| <b>OBD selection (%)</b>                             |        |        |        |        |        |          |
| Oracle   one-stage   cycle 1, alpha=0                | 15.2   | 24.2   | 42.6   | 10.5   | 0.8    | 6.7      |
| New non-TITE   one-stage   alpha=0                   | 16.4   | 22.3   | 43.7   | 10.9   | 1.1    | 5.6      |
| New non-TITE   one-stage   alpha=0.3                 | 15.8   | 23.9   | 44.7   | 8.7    | 0.5    | 6.4      |
| New non-TITE   one-stage   alpha=0.6                 | 14.3   | 28.2   | 44.6   | 5.6    | 0.2    | 7.1      |
| New non-TITE   one-stage   alpha=0.9                 | 16.1   | 32.8   | 39.6   | 3.1    | 0      | 8.4      |
| Oracle   2-stage highest utility   cycle 1, alpha=0  | 16.4   | 25.1   | 43.9   | 6.8    | 0.6    | 7.2      |
| New non-TITE   2-stage highest utility   alpha=0     | 16.3   | 23.9   | 44     | 8.4    | 0.9    | 6.5      |
| New non-TITE   2-stage highest utility   alpha=0.3   | 17.8   | 27.6   | 41.5   | 5.4    | 0.4    | 7.3      |
| New non-TITE   2-stage highest utility   alpha=0.6   | 19.9   | 30.1   | 37.8   | 4.5    | 0.2    | 7.5      |
| New non-TITE   2-stage highest utility   alpha=0.9   | 22.1   | 34     | 32.4   | 3      | 0      | 8.5      |
| Oracle   2-stage top-2 randomized   cycle 1, alpha=0 | 12     | 24.5   | 45.2   | 10.6   | 1.1    | 6.6      |
| New non-TITE   2-stage top-2 randomized   alpha=0    | 10.5   | 23.6   | 46.6   | 12.5   | 1.1    | 5.7      |
| New non-TITE   2-stage top-2 randomized   alpha=0.3  | 11.3   | 28.7   | 41.5   | 10.7   | 0.8    | 7        |
| New non-TITE   2-stage top-2 randomized   alpha=0.6  | 13.4   | 31.5   | 39.8   | 6.9    | 0.4    | 8        |
| New non-TITE   2-stage top-2 randomized   alpha=0.9  | 17.2   | 31.3   | 37.2   | 5.2    | 0.2    | 8.9      |
| <b>Mean administrations by dose</b>                  |        |        |        |        |        |          |
| Oracle   one-stage   cycle 1, alpha=0                | 7.3    | 8.1    | 9.7    | 3.4    | 0.6    |          |
| New non-TITE   one-stage   alpha=0                   | 7.1    | 7.9    | 9.8    | 3.6    | 0.7    |          |
| New non-TITE   one-stage   alpha=0.3                 | 7.6    | 8.4    | 9.7    | 3      | 0.4    |          |
| New non-TITE   one-stage   alpha=0.6                 | 7.9    | 8.8    | 9.6    | 2.4    | 0.3    |          |
| New non-TITE   one-stage   alpha=0.9                 | 8.4    | 9.5    | 8.9    | 1.9    | 0.1    |          |
| Oracle   2-stage highest utility   cycle 1, alpha=0  | 7.8    | 8.2    | 9.4    | 3      | 0.7    |          |
| New non-TITE   2-stage highest utility   alpha=0     | 7.7    | 7.9    | 9.4    | 3.2    | 0.9    |          |
| New non-TITE   2-stage highest utility   alpha=0.3   | 8.3    | 8.3    | 8.9    | 2.8    | 0.7    |          |
| New non-TITE   2-stage highest utility   alpha=0.6   | 8.7    | 8.8    | 8.6    | 2.5    | 0.5    |          |
| New non-TITE   2-stage highest utility   alpha=0.9   | 9.3    | 9.3    | 7.9    | 2.1    | 0.4    |          |
| Oracle   2-stage top-2 randomized   cycle 1, alpha=0 | 7.6    | 8.4    | 8.5    | 3.7    | 0.9    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0    | 7.4    | 8.2    | 8.4    | 4      | 1.1    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.3  | 8      | 8.6    | 8.1    | 3.6    | 0.8    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.6  | 8.5    | 8.9    | 7.9    | 3.1    | 0.6    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.9  | 9.3    | 9.2    | 7.5    | 2.5    | 0.4    |          |

# Phase I/II Operating Characteristics

New non-TITE alpha comparison with cycle-1 oracle (alpha = 0)

|                                                      | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | No OBD % |
|------------------------------------------------------|--------|--------|--------|--------|--------|----------|
| <b>Scenario 38</b>                                   |        |        |        |        |        |          |
| DLT rate                                             | 0.05   | 0.1    | 0.12   | 0.15   | 0.3    |          |
| Efficacy rate                                        | 0.05   | 0.08   | 0.12   | 0.15   | 0.5    |          |
| <b>OBD selection (%)</b>                             |        |        |        |        |        |          |
| Oracle   one-stage   cycle 1, alpha=0                | 13.9   | 7.6    | 6.9    | 7.1    | 63.1   | 1.4      |
| New non-TITE   one-stage   alpha=0                   | 14     | 7.9    | 6      | 7.8    | 63.2   | 1.1      |
| New non-TITE   one-stage   alpha=0.3                 | 14.7   | 7.7    | 7.8    | 10.4   | 58     | 1.4      |
| New non-TITE   one-stage   alpha=0.6                 | 14.9   | 9.2    | 8.6    | 11.9   | 53.8   | 1.6      |
| New non-TITE   one-stage   alpha=0.9                 | 15.9   | 9.7    | 9.7    | 12.4   | 50.6   | 1.7      |
| Oracle   2-stage highest utility   cycle 1, alpha=0  | 19.2   | 10.3   | 8.6    | 7.4    | 52.8   | 1.7      |
| New non-TITE   2-stage highest utility   alpha=0     | 17.9   | 10     | 7.3    | 7.5    | 56     | 1.3      |
| New non-TITE   2-stage highest utility   alpha=0.3   | 18.5   | 10.7   | 8.9    | 10     | 50     | 1.9      |
| New non-TITE   2-stage highest utility   alpha=0.6   | 20.3   | 12.4   | 8.7    | 10.7   | 45.7   | 2.2      |
| New non-TITE   2-stage highest utility   alpha=0.9   | 20.6   | 12.6   | 12     | 12.1   | 40.5   | 2.2      |
| Oracle   2-stage top-2 randomized   cycle 1, alpha=0 | 12     | 5.6    | 7.4    | 7.8    | 65.8   | 1.4      |
| New non-TITE   2-stage top-2 randomized   alpha=0    | 11.7   | 6      | 5.5    | 8.5    | 67.3   | 1        |
| New non-TITE   2-stage top-2 randomized   alpha=0.3  | 12.3   | 7.5    | 8.3    | 10.6   | 59.8   | 1.5      |
| New non-TITE   2-stage top-2 randomized   alpha=0.6  | 12.8   | 8.6    | 8.2    | 11.2   | 57     | 2.2      |
| New non-TITE   2-stage top-2 randomized   alpha=0.9  | 11.9   | 10     | 11.7   | 11.1   | 53.2   | 2.1      |
| <b>Mean administrations by dose</b>                  |        |        |        |        |        |          |
| Oracle   one-stage   cycle 1, alpha=0                | 4.6    | 4.6    | 5.1    | 5.3    | 10.2   |          |
| New non-TITE   one-stage   alpha=0                   | 4.5    | 4.5    | 5      | 5.3    | 10.4   |          |
| New non-TITE   one-stage   alpha=0.3                 | 4.8    | 4.8    | 5.2    | 5.5    | 9.5    |          |
| New non-TITE   one-stage   alpha=0.6                 | 4.8    | 5      | 5.4    | 5.6    | 8.9    |          |
| New non-TITE   one-stage   alpha=0.9                 | 4.9    | 5.4    | 5.5    | 5.6    | 8.3    |          |
| Oracle   2-stage highest utility   cycle 1, alpha=0  | 4.9    | 4.8    | 5.5    | 5.6    | 9.1    |          |
| New non-TITE   2-stage highest utility   alpha=0     | 4.8    | 4.7    | 5.3    | 5.4    | 9.6    |          |
| New non-TITE   2-stage highest utility   alpha=0.3   | 5      | 4.9    | 5.6    | 5.7    | 8.6    |          |
| New non-TITE   2-stage highest utility   alpha=0.6   | 5.1    | 5.2    | 5.8    | 5.7    | 8      |          |
| New non-TITE   2-stage highest utility   alpha=0.9   | 5.2    | 5.6    | 6      | 5.7    | 7.2    |          |
| Oracle   2-stage top-2 randomized   cycle 1, alpha=0 | 4.8    | 4.9    | 5.4    | 6.8    | 7.8    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0    | 4.7    | 4.8    | 5.3    | 6.9    | 8.2    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.3  | 4.8    | 5      | 5.5    | 6.8    | 7.5    |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.6  | 4.9    | 5.4    | 5.7    | 6.7    | 7      |          |
| New non-TITE   2-stage top-2 randomized   alpha=0.9  | 5      | 5.7    | 5.9    | 6.6    | 6.5    |          |